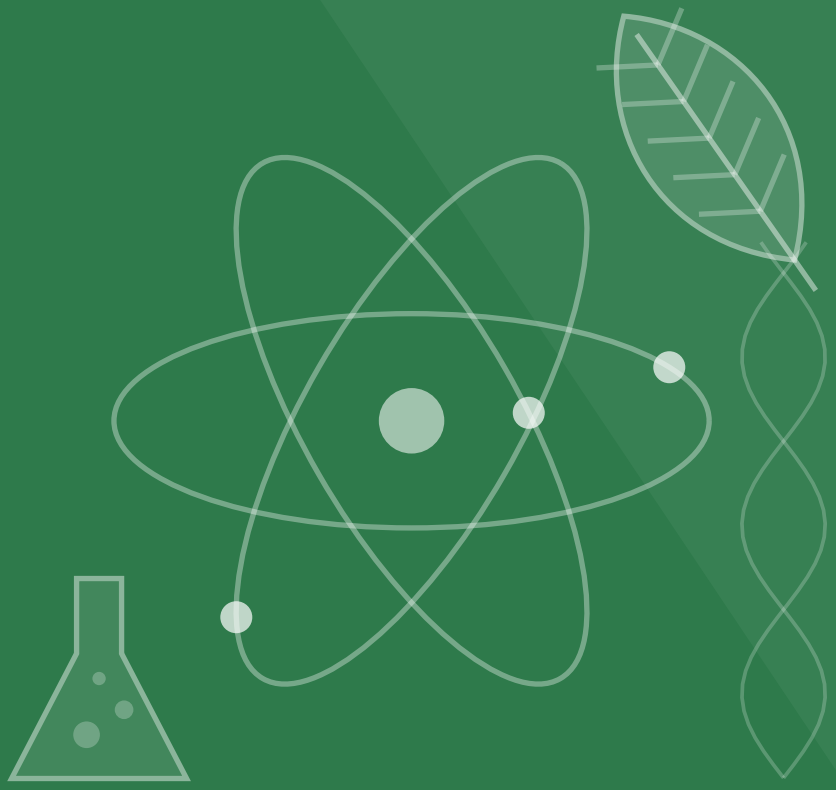
VENVLEARNING

TEACHER EDITION



Science

GRADE 4

Guided lessons, pacing, and complete answer keys

Unit 1: The Human Body

VENVLEARNING

Grade 4
Science

TEACHER EDITION

Unit 1 The Human Body

A VenvLearning family homeschool curriculum

How to use this Teacher Edition

This book holds everything needed to teach the course: unit overviews, a plan for every lesson with what to say and do, the student lesson with Try It answers, and answer keys for every workbook page, weekly check and unit test.

- Daily routine (about 45–55 minutes): Engage 5 · Investigate 15 · Explain 10 · Guided Practice 7 · Independent 10 · Exit Ticket 3.
- Say: lines are a suggested script. Use your own words.
- Two lessons a week, then the Weekly Check (about 15 minutes), then corrections or a project.
- Each plan's Pacing row and each unit's pacing guide give the week and lesson order. Set the days to fit your own calendar.

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UNIT 1 · TEACHER OVERVIEW

The Human Body

Thinking like a scientist; cells, tissues, and organs; and the body systems that keep you alive.

Your body is an amazing machine made of trillions of cells. In this unit you'll learn to think like a scientist: observing, asking testable questions, running fair tests, and using evidence. Then you'll use those skills to explore how your body is organized and how its systems work together.

Big ideas

- Tell observations from inferences, and plan fair tests.
- Support claims with evidence and reasoning (CER).
- Describe levels of organization from cells to systems.
- Explain the jobs of the digestive, respiratory, circulatory, skeletal, muscular, and nervous systems and how they work together.

Standards

Code	Standard
4-LS1-1	Internal and external structures support survival, growth, behavior, and reproduction.
SEP 1, 3, 4, 7	Ask questions; plan and carry out investigations; analyze data; argue from evidence.

What students should already know

- Living things need food, water, and air.

Materials for the unit

- Paper bags and small objects
- Paper towels, cups, measuring cup
- Building blocks
- Zip bags, crackers, vinegar
- Timer

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 2	1.1	Thinking Like a Scientist
Week 1	Lesson 2 of 2	1.2	Fair Tests and Evidence
Week 1	End of week	Check	Weekly Check 1: Thinking Like a Scientist
Week 2	Lesson 1 of 2	1.3	Cells, Tissues, Organs, and Systems
Week 2	Lesson 2 of 2	1.4	Body Systems Work Together
Week 2	End of week	Check	Weekly Check 2: Body Systems
Week 3	Lesson 1 of 2	1.5	The Digestive System
Week 3	Lesson 2 of 2	1.6	The Respiratory System
Week 3	End of week	Check	Weekly Check 3: Digestion and Breathing
Week 4	Lesson 1 of 2	1.7	The Heart and Blood Vessels
Week 4	Lesson 2 of 2	1.8	Bones, Muscles, and Healthy Choices
Week 4	Before the test	Review	Unit 1 Review
Week 4	End of unit	Test	Unit 1 Test

TEACHING THIS UNIT AT HOME

Take pulses and count breaths as a family before and after a walk, and compare the results.

UNIT VOCABULARY

observation	Information gathered with senses or tools.
inference	A conclusion based on observations.
fair test	Only one variable is changed.
cell	The smallest living unit.
organ system	Organs that work together.
nutrients	Substances in food the body uses.

LESSON 1.1 · TEACHER PLAN
Week 1 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.1: Thinking Like a Scientist

Student goal	I can tell the difference between an observation and an inference, and ask a question that can be investigated.
Objective	Students distinguish observations from inferences and write testable questions.
Standard	Science and Engineering Practices 1, 3 — Ask questions; plan and carry out investigations.
Pacing	Unit 1, week 1, lesson 1 of 2
Time	About 50 minutes
Materials	Paper bags with hidden objects, workbook Lesson 1.1
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Cookie jar story: see vs think.
Investigate	15	Mystery Bag investigation.
Explain	10	Chart observations vs inferences. Model a testable question.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating inferences as facts.

SUPPORT

- Sentence frames: I observe… / I infer…

EXTEND

- Observe a pet or bird for 10 minutes and record observations.

STUDENT LESSON 1.1 · WITH ANSWERS

Lesson 1.1: Thinking Like a Scientist

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Do the Investigate activity: Mystery Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–D).

I WONDER

You walk into the kitchen and see crumbs on the counter and an empty cookie jar. What do you SEE? What do you THINK happened?

Scientists are curious people who ask questions about the world and then look for evidence to answer them. Two of their most important tools are observations and inferences.

Observation	Inference
what you notice with your senses or tools	what you figure out from what you noticed
“The cookie jar is empty.”	“Someone ate the cookies.”
“The ground is wet.”	“It rained last night.”
“The thermometer reads 38°F.”	“It is too cold to swim outside.”

An inference can be wrong. Maybe the cookies were moved to another jar! Good scientists gather more observations to test their inferences.

EXPLAIN IT · ASK A TESTABLE QUESTION

You notice that a puddle in the sun dries faster than a puddle in the shade.

1 Observation: the sunny puddle disappeared first.

2 Question that can't be tested easily: “Why does water exist?”

3 Testable question: “Does water evaporate faster in sunlight than in shade?” You can set up two cups of water and measure.

Explanation: A testable question can be answered by observing, measuring, or experimenting.

INVESTIGATE · MYSTERY BAG

What can you figure out about an object you can't see?

You need: a paper bag with a hidden object (chosen by your teacher); paper and a pencil.

1 Without looking, shake the bag and listen. Feel the object through the paper.

2 List at least four observations (sound, weight, shape, texture).

3 Write one inference about what the object is, and the observations that support it.

4 Open the bag. Was your inference correct?

Record: a two-column chart: Observations | Inferences.

★ KEY IDEA

Observations come from your senses and tools. Inferences are explanations based on observations. Testable questions can be answered with evidence.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Observation or inference: “The leaf is green and has jagged edges.”

Answer: observation

2. Observation or inference: “The dog is hungry because it is barking at its bowl.”

Answer: inference

3. Make this question testable: “Why do plants like light?”

Answer: e.g. Does a bean plant grow taller with more hours of light?

Workbook Answer Key · Lesson 1.1

#	Question	Answer
1	Observation or inference: “The leaf is green and has jagged edges.”	observation
2	Observation or inference: “The dog is hungry because it is barking at its bowl.”	inference
3	Make this question testable: “Why do plants like light?”	e.g. Does a bean plant grow taller with more hours of light?
4	Name two of your senses.	e.g. sight, hearing, touch, smell, taste
5	Name one tool that helps scientists observe.	e.g. magnifying glass, ruler, thermometer
6	“The water is 20°C.” Observation or inference?	observation

#	Question	Answer
7	“The ice melted because the room is warm.” Observation or inference?	inference
8	“The rock feels rough.” Observation or inference?	observation
9	“A bird built this nest.” Observation or inference?	inference
10	Look out a window. Write two observations and one inference.	Answers vary; observations use senses, the inference explains them.
11	Write a testable question about something in your home.	Answers vary; must be answerable by observing or measuring.
12	In your own words, what is the difference between an observation and an inference?	An observation is what you notice with your senses or tools; an inference is what you conclude from it.

LESSON 1.2 · TEACHER PLAN
Week 1 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.2: Fair Tests and Evidence

Student goal	I can plan a fair test and use evidence to support a claim.
Objective	Students plan fair tests and write CER.
Standard	Science and Engineering Practices 3, 4, 7 — Plan and carry out fair tests; construct explanations from evidence.
Pacing	Unit 1, week 1, lesson 2 of 2
Time	About 50 minutes
Materials	Paper towels, measuring cups, water, workbook Lesson 1.2
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Paper airplane race.
Investigate	15	Paper Towel Test.
Explain	10	Teach variables; CER.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Changing more than one variable.

SUPPORT

- Change/Same/Measure chart.

EXTEND

- Repeat the test three times and average the results.

STUDENT LESSON 1.2 · WITH ANSWERS

Lesson 1.2: Fair Tests and Evidence

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Do the Investigate activity: The Paper Towel Test.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–D).

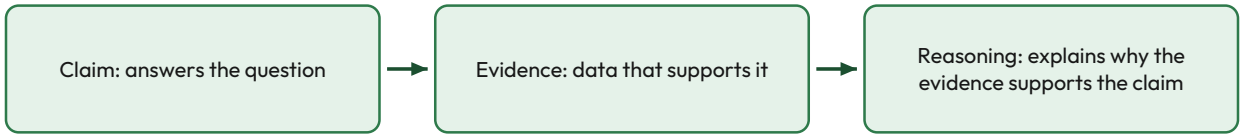
I WONDER

Two friends race paper airplanes. One throws hard and one throws gently. Is it a fair race? Why or why not?

In a fair test, you change only one thing, the variable you test, and keep everything else the same. Then you measure what happens. If you change two things at once, you can't tell which one caused the result.

Question	Change (test)	Keep the same	Measure
Which paper towel brand absorbs the most water?	brand of towel	towel size, amount of water, time	water left in the cup (mL)

After a test, scientists write a claim, support it with evidence (data), and give reasoning that explains why the evidence supports the claim. This is called CER.



INVESTIGATE · THE PAPER TOWEL TEST

Which paper towel absorbs the most water?

You need: 2 or 3 kinds of paper towel cut to the same size; measuring cup; water; 3 cups.

1. Pour 100 mL of water into each cup.
2. Dip one towel square in each cup for 10 seconds.
3. Lift the towel and let it drip back into the cup for 10 seconds.
4. Measure the water left in each cup.

Record: a data table, then a CER: Which towel absorbed the most? What is your evidence?

★ KEY IDEA

Change one variable, keep the rest the same, measure the result, and support your claim with evidence and reasoning.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Why must the towel squares be the same size?

Answer: So that only the brand changes; size would also affect how much water is absorbed.

2. What is the claim in CER?

Answer: a statement that answers the question

3. A student tests plant growth but gives one plant more water AND more light. What is wrong?

Answer: Two variables changed, so you can't tell which caused the result.

Workbook Answer Key · Lesson 1.2

#	Question	Answer
1	Why must the towel squares be the same size?	So that only the brand changes; size would also affect how much water is absorbed.
2	What is the claim in CER?	a statement that answers the question
3	A student tests plant growth but gives one plant more water AND more light. What is wrong?	Two variables changed, so you can't tell which caused the result.
4	What does a fair test change?	only one variable
5	What do you call data that supports a claim?	evidence
6	You want to know if a ball bounces higher on carpet or wood. What will you change? What will you keep the same? What will you measure?	Change: floor surface. Same: ball, drop height. Measure: bounce height.
7	Brand A towel left 40 mL; Brand B left 65 mL; Brand C left 52 mL (from 100 mL). Which absorbed the most?	Brand A (it absorbed 60 mL)
8	Write a CER for the towel data above.	Claim: Brand A absorbs the most. Evidence: only 40 mL was left (60 mL absorbed) vs 35 and 48 mL. Reasoning: less water left means more was soaked up.
9	Why do scientists keep all variables the same except one?	So they know the one change caused the result.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.2

About 15 minutes

Weekly Check 1: Thinking Like a Scientist — Answer Key

Answers

#	Question	Answer
1	Observation or inference: “The sky is cloudy.”	observation
2	Observation or inference: “It will probably rain.”	inference
3	Make this testable: “Why are some balls bouncy?”	e.g. Which ball bounces highest when dropped from 1 meter?
4	A student tests whether salt water or fresh water freezes faster, but puts one cup in the freezer and one in the refrigerator. What is wrong?	Two variables changed (type of water and location), so it isn't a fair test.
5	What does the E in CER stand for?	evidence

LESSON 1.3 · TEACHER PLAN
Week 2 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.3: Cells, Tissues, Organs, and Systems

Student goal	I can describe how the body is organized from cells to organ systems.
Objective	Students describe levels of organization.
Standard	4-LS1-1 — Internal structures support survival and function.
Pacing	Unit 1, week 2, lesson 1 of 2
Time	About 50 minutes
Materials	Building blocks, sticky notes, workbook Lesson 1.3
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Brick → wall → room → house.
Investigate	15	Build the Levels.
Explain	10	Teach the flow chart; cell shapes fit jobs.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking organs are made of only one kind of cell.

SUPPORT

- Flow chart with pictures.

EXTEND

- Look at onion skin or a leaf with a magnifier or microscope.

STUDENT LESSON 1.3 · WITH ANSWERS

Lesson 1.3: Cells, Tissues, Organs, and Systems

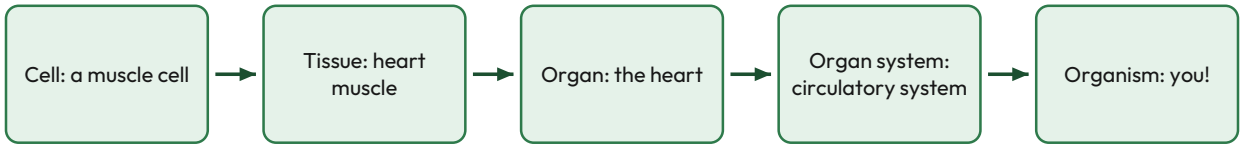
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Do the Investigate activity: Build the Levels.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–D).

I WONDER

A brick is part of a wall, a wall is part of a room, and a room is part of a house. How might your body be built the same way?

Your body is made of trillions of tiny cells, so small you need a microscope to see them. Similar cells work together as tissues, such as muscle tissue. Different tissues form organs, like the heart or stomach. Organs work together in organ systems.



There are many kinds of cells. Nerve cells are long and thin to carry messages. Red blood cells are round and flat to carry oxygen. A cell's shape fits its job.

INVESTIGATE · BUILD THE LEVELS

How can a model show levels of organization?

You need: building blocks or LEGO bricks; sticky notes.

1. Let one brick stand for one cell.
2. Join several matching bricks into a "tissue."
3. Combine different "tissues" into an "organ."
4. Put several organs together into a "system." Label each level with a sticky note.

Record: a sketch of your model with each level labeled.

★ KEY IDEA

Cells → tissues → organs → organ systems → organism.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What is the smallest living unit of the body?

Answer: a cell

2. Put in order: organ, cell, organ system, tissue

Answer: cell, tissue, organ, organ system

3. Name one organ.

Answer: e.g. heart, lungs, stomach, brain, skin

Workbook Answer Key · Lesson 1.3

#	Question	Answer
1	What is the smallest living unit of the body?	a cell
2	Put in order: organ, cell, organ system, tissue	cell, tissue, organ, organ system
3	Name one organ.	e.g. heart, lungs, stomach, brain, skin
4	A group of similar cells is a…	tissue
5	The heart is an…	organ
6	Smallest level of organization	cell
7	Organs working together form an…	organ system
8	Why are nerve cells long and thin?	to carry messages over long distances
9	Is skin an organ?	yes (the largest organ)
10	Use the brick-and-house comparison to explain levels of organization.	Cells are like bricks; tissues like walls; organs like rooms; systems like the whole house.
11	Draw and label the four levels, from cell to organ system.	cell, tissue, organ, organ system (e.g. muscle cell, muscle tissue, heart, circulatory system)

LESSON 1.4 · TEACHER PLAN
Week 2 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.4: Body Systems Work Together

Student goal	I can name major body systems, describe each one's main job, and explain how they work together.
Objective	Students identify body systems and their interactions.
Standard	4-LS1-1 — Internal structures function to support survival, growth, and behavior.
Pacing	Unit 1, week 2, lesson 2 of 2
Time	About 50 minutes
Materials	System index cards, workbook Lesson 1.4
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Soccer kick.
Investigate	15	Systems Relay.
Explain	10	Teach the systems chart. Model the soccer example.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking each system works alone.

SUPPORT

- Picture cards for each system.

EXTEND

- Make a body-systems poster.

STUDENT LESSON 1.4 · WITH ANSWERS

Lesson 1.4: Body Systems Work Together

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Do the Investigate activity: Systems Relay.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–E).

I WONDER

Think about kicking a soccer ball. Which parts of your body are working?

System	Main organs	Main job
skeletal	bones	supports, protects, works with muscles to move
muscular	muscles	moves bones and organs
nervous	brain, spinal cord, nerves	senses, thinks, sends messages
digestive	mouth, stomach, intestines	breaks food into nutrients
respiratory	nose, windpipe, lungs	takes in oxygen, gets rid of carbon dioxide
circulatory	heart, blood vessels, blood	carries oxygen and nutrients to cells

EXPLAIN IT · SYSTEMS WORKING TOGETHER

Kicking a soccer ball

1 Nervous: your eyes see the ball and your brain sends a message to your leg.
2 Muscular and skeletal: leg muscles pull on bones to swing your leg.
3 Respiratory and circulatory: you breathe faster and your heart pumps faster to deliver oxygen to your muscles.
4 Digestive: nutrients from food give your cells energy.

Explanation: No system works alone; all of them work together to keep you alive and moving.

INVESTIGATE · SYSTEMS RELAY

Which systems work when you do an everyday task?

You need: index cards labeled with the six systems; a timer.

1 Choose an action: eating an apple, running up stairs, or reading a book aloud.

2 Do the action (safely!).

3 Hold up every system card you think was working, and explain why.

Record: the action and the systems used, with one reason for each.

★ KEY IDEA

Each body system has a main job, and the systems depend on one another.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which system carries oxygen to cells?

Answer: circulatory

2. Which system breaks down food?

Answer: digestive

3. Which systems help you jump?

Answer: nervous, muscular, skeletal (also respiratory, circulatory)

Workbook Answer Key · Lesson 1.4

#	Question	Answer
1	Which system carries oxygen to cells?	circulatory
2	Which system breaks down food?	digestive
3	Which systems help you jump?	nervous, muscular, skeletal (also respiratory, circulatory)
4	The brain belongs to the ___ system.	nervous
5	Bones belong to the ___ system.	skeletal
6	lungs	respiratory
7	heart	circulatory
8	stomach	digestive
9	spinal cord	nervous
10	skull	skeletal
11	biceps	muscular

#	Question	Answer
12	Choose an activity (swimming, playing piano, eating lunch). Explain how three systems work together.	Answers vary; three systems with jobs.
13	Why do you breathe faster when you run?	Muscles need more oxygen; the respiratory and circulatory systems work harder to deliver it.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.4

About 15 minutes

Weekly Check 2: Body Systems — Answer Key

Answers

#	Question	Answer
1	Put in order: organ, cell, tissue, organ system	cell, tissue, organ, organ system
2	Which system includes the brain?	nervous
3	Which system breaks down food?	digestive
4	Which system moves blood?	circulatory
5	Name two systems that work together when you climb stairs and explain how.	e.g. muscular and skeletal: muscles pull bones to lift legs.

LESSON 1.5 · TEACHER PLAN
Week 3 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.5: The Digestive System

Student goal	I can describe the path of food through the digestive system and explain what each organ does.
Objective	Students trace food through the digestive system.
Standard	4-LS1-1 — Internal structures support survival.
Pacing	Unit 1, week 3, lesson 1 of 2
Time	About 50 minutes
Materials	Zip bags, crackers, water, vinegar, workbook Lesson 1.5
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Where does breakfast go?
Investigate	15	Stomach in a Bag.
Explain	10	Teach the path; mechanical vs chemical digestion.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking food goes straight to the stomach and stops.

SUPPORT

- Trace the path on a body outline.

EXTEND

- Measure 20 feet of string to see the small intestine's length.

STUDENT LESSON 1.5 · WITH ANSWERS

Lesson 1.5: The Digestive System

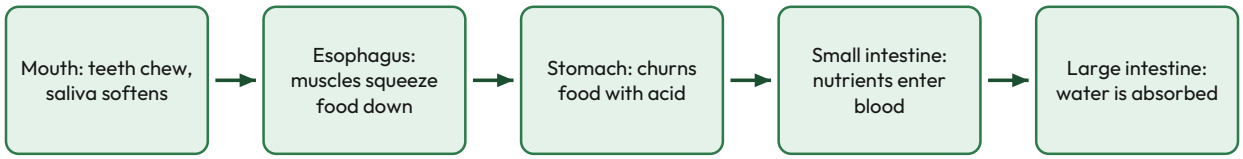
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Do the Investigate activity: Stomach in a Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–D).

I WONDER

Where does your breakfast go after you swallow it?

Your body can't use a sandwich as a sandwich. The digestive system breaks food into tiny nutrients that can pass into your blood and travel to your cells.



Digestion is both mechanical (chewing and churning break food into smaller pieces) and chemical (saliva, stomach acid, and other juices break food down into nutrients). The liver and pancreas make juices that help in the small intestine. In an adult, the small intestine is about 20 feet long, coiled up to fit inside the body!

INVESTIGATE · STOMACH IN A BAG

How does the stomach help break down food?

You need: zip-top plastic bag; 4 crackers; 1/2 cup water; a few drops of vinegar (to stand for acid).

1. Put the crackers in the bag. Observe them.
2. Add the water and vinegar and seal the bag tightly.
3. Gently squeeze and mash the bag for 2 minutes, the way stomach muscles churn.
4. Observe the crackers again.

Record: before-and-after observations, and which part of digestion the squeezing models.

Safety: Do not eat or drink anything used in an investigation.

★ KEY IDEA

Food travels mouth → esophagus → stomach → small intestine → large intestine. Nutrients enter the blood in the small intestine.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Where do most nutrients enter the blood?

Answer: the small intestine

2. What does the large intestine absorb?

Answer: water

3. What did squeezing the bag model?

Answer: the stomach churning food (mechanical digestion)

Workbook Answer Key · Lesson 1.5

#	Question	Answer
1	Where do most nutrients enter the blood?	the small intestine
2	What does the large intestine absorb?	water
3	What did squeezing the bag model?	the stomach churning food (mechanical digestion)
4	What do teeth do to food?	chew/break it into smaller pieces
5	What are nutrients?	substances in food the body uses
6	Put in order: stomach, mouth, large intestine, esophagus, small intestine	mouth, esophagus, stomach, small intestine, large intestine
7	Mechanical or chemical: teeth chewing	mechanical
8	Mechanical or chemical: stomach acid	chemical
9	Which organ is a long, coiled tube where nutrients enter the blood? A. esophagus B. stomach C. small intestine D. large intestine	C. small intestine
10	Explain why food must be broken down before the body can use it.	Cells can only use tiny nutrients that pass into the blood, not whole pieces of food.
11	Describe what happened to the crackers in the bag and what it shows about digestion.	They became mushy and broke apart; the stomach mixes and breaks down food.

LESSON 1.6 · TEACHER PLAN
Week 3 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.6: The Respiratory System

Student goal	I can explain how the respiratory system brings oxygen into the body and removes carbon dioxide.
Objective	Students explain gas exchange and breathing.
Standard	4-LS1-1 — Internal structures support survival.
Pacing	Unit 1, week 3, lesson 2 of 2
Time	About 50 minutes
Materials	Timer, workbook Lesson 1.6
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Hold your breath.
Investigate	15	Breathing Rate investigation.
Explain	10	Teach the path and the diaphragm.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking we breathe out no oxygen at all, or that lungs are muscles.

SUPPORT

- Hands on ribs while breathing.

EXTEND

- Build a balloon lung model in a plastic bottle.

STUDENT LESSON 1.6 · WITH ANSWERS

Lesson 1.6: The Respiratory System

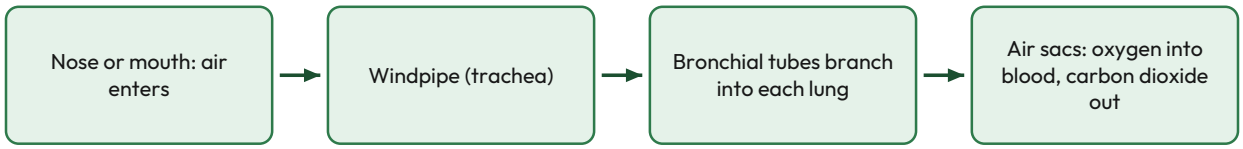
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Do the Investigate activity: Breathing Rate.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–E).

I WONDER

Hold your breath for a few seconds. Why can't you hold it forever?

Every cell in your body needs oxygen to release energy from nutrients. Cells also make carbon dioxide, a waste gas. The respiratory system brings oxygen in and sends carbon dioxide out.



When the diaphragm tightens and moves down, the chest gets bigger and air rushes into the lungs. When it relaxes and moves up, air is pushed out. Inside the lungs are millions of tiny air sacs wrapped in blood vessels, where oxygen moves into the blood and carbon dioxide moves out.

INVESTIGATE · BREATHING RATE

How does exercise change how fast you breathe?

You need: a timer; a partner.

1. Sit quietly for 1 minute. Count your breaths for 30 seconds and multiply by 2.
2. Do jumping jacks for 1 minute.
3. Right away, count your breaths for 30 seconds and multiply by 2.
4. Rest 3 minutes and count again.

Record: breaths per minute at rest, after exercise, and after resting, in a table and a bar graph.

★ KEY IDEA

You breathe in oxygen that cells need and breathe out carbon dioxide that cells make. Gases are exchanged in the air sacs.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What gas do your cells need?

Answer: oxygen

2. What muscle helps you breathe?

Answer: the diaphragm

3. Why does breathing speed up during exercise?

Answer: Muscles need more oxygen and make more carbon dioxide.

Workbook Answer Key · Lesson 1.6

#	Question	Answer
1	What gas do your cells need?	oxygen
2	What muscle helps you breathe?	the diaphragm
3	Why does breathing speed up during exercise?	Muscles need more oxygen and make more carbon dioxide.
4	The gas you breathe out is...	carbon dioxide
5	The windpipe is also called the...	trachea
6	Order: air sacs, nose, windpipe, bronchial tubes	nose, windpipe, bronchial tubes, air sacs
7	Where does oxygen pass into the blood?	in the air sacs of the lungs
8	Maya breathed 18 times per minute at rest and 36 after exercise. How many more?	18 more (twice as many)
9	Explain how the respiratory and circulatory systems work together.	The lungs put oxygen into the blood; the heart pumps that blood to cells and brings carbon dioxide back to the lungs.
10	Make a bar graph of your breathing data: at rest, after exercise, after resting.	Bars labeled with breaths per minute; after-exercise bar is tallest.
11	What happens when the diaphragm moves down?	The chest gets bigger and air flows into the lungs.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.6

About 15 minutes

Weekly Check 3: Digestion and Breathing — Answer Key

Answers

#	Question	Answer
1	Where do most nutrients enter the blood?	small intestine
2	What does the large intestine absorb?	water
3	What gas do you breathe out?	carbon dioxide
4	Where are gases exchanged in the lungs?	air sacs
5	Which is mechanical digestion? A. stomach acid B. chewing C. saliva chemicals	B. chewing
6	Ben breathes 16 times a minute at rest and 34 after running. Explain why.	Running muscles need more oxygen and produce more carbon dioxide, so he breathes faster.

LESSON 1.7 · TEACHER PLAN
Week 4 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.7: The Heart and Blood Vessels

Student goal	I can describe how the heart and blood vessels move blood through the body, and measure my pulse.
Objective	Students describe circulation and measure pulse.
Standard	4-LS1-1 — Internal structures support survival.
Pacing	Unit 1, week 4, lesson 1 of 2
Time	About 50 minutes
Materials	Timer, workbook Lesson 1.7
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Fist-size heart.
Investigate	15	Take Your Pulse.
Explain	10	Teach vessels; the cycle diagram.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking blood in veins is blue.

SUPPORT

- Body outline with red and blue arrows labeled “to heart” / “from heart.”

EXTEND

- Calculate how many times your heart beats in a day.

STUDENT LESSON 1.7 · WITH ANSWERS

Lesson 1.7: The Heart and Blood Vessels

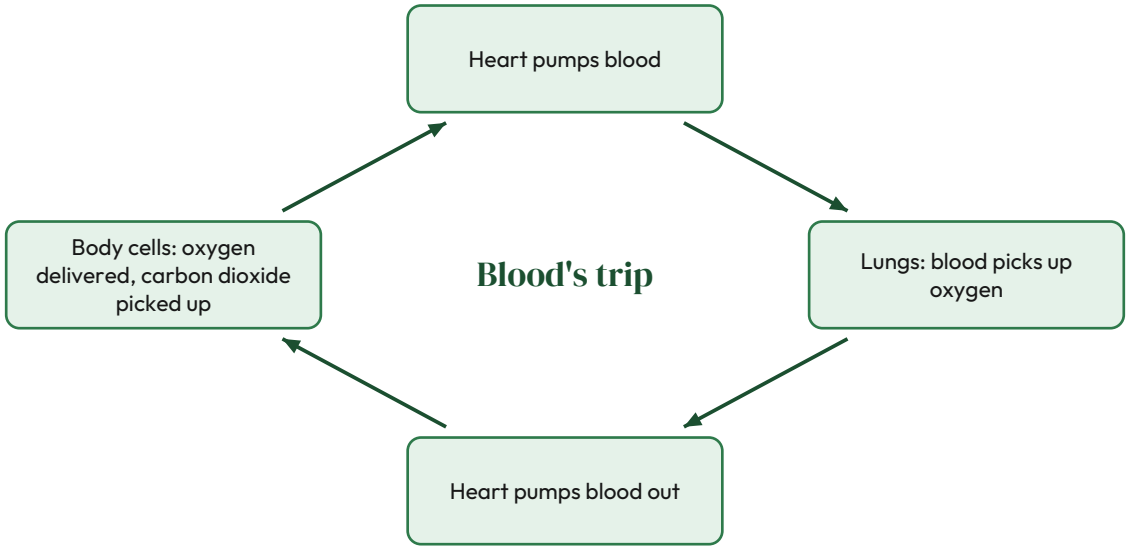
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Do the Investigate activity: Take Your Pulse.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–D).

I WONDER

Make a fist. Your heart is about that size. How many times do you think it beats in one minute?

The heart is a muscle about the size of your fist. It has four chambers and pumps blood through a huge network of blood vessels. Arteries carry blood away from the heart. Capillaries are tiny vessels where oxygen and nutrients move into cells and wastes move out. Veins carry blood back to the heart.



INVESTIGATE · TAKE YOUR PULSE

How does activity change heart rate?

You need: a timer.

1. Place two fingers (not your thumb) on the side of your neck or the inside of your wrist.
2. Count the beats for 15 seconds. Multiply by 4 to find beats per minute.
3. Run in place for 1 minute, then measure again.

4 Rest 3 minutes and measure again.

Record: beats per minute in a table; compare with your breathing data from Lesson 1.6.

! WATCH OUT

Blood in veins is not blue. Blood is always red; veins can look bluish through skin because of how light passes through it.

★ KEY IDEA

The heart pumps blood through arteries to capillaries and back through veins, delivering oxygen and nutrients and carrying away wastes.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which vessels carry blood away from the heart?

Answer: arteries

2. Where do oxygen and nutrients move into cells?

Answer: capillaries

3. 15-second count is 22 beats. Beats per minute?

Answer: 88

Workbook Answer Key · Lesson 1.7

#	Question	Answer
1	Which vessels carry blood away from the heart?	arteries
2	Where do oxygen and nutrients move into cells?	capillaries
3	15-second count is 22 beats. Beats per minute?	88
4	The heart is about the size of your…	fist
5	Veins carry blood…	back to the heart
6	Pulse count of 20 in 15 seconds = ___ beats per minute	80
7	Pulse count of 30 in 15 seconds = ___ beats per minute	120
8	Why does heart rate go up during exercise?	Muscles need more oxygen and nutrients, so blood must move faster.
9	Which vessel is the tiniest? A. artery B. vein C. capillary	C. capillary

#	Question	Answer
10	Trace a drop of blood from the heart to the lungs and back to the body.	Heart → lungs (pick up oxygen) → heart → body cells (deliver oxygen) → heart.
11	Why do the respiratory and circulatory systems both speed up during exercise?	Muscles need more oxygen; lungs take in more and the heart delivers it faster.

LESSON 1.8 · TEACHER PLAN
Week 4 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.8: Bones, Muscles, and Healthy Choices

Student goal	I can explain how bones and muscles work together and describe choices that keep body systems healthy.
Objective	Students explain bone-muscle teamwork and healthy habits.
Standard	4-LS1-1 — Internal structures support survival, growth, and behavior.
Pacing	Unit 1, week 4, lesson 2 of 2
Time	About 50 minutes
Materials	Workbook Lesson 1.8, Unit 1 Review
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Bend your arm.
Investigate	15	Muscle Partners.
Explain	10	Teach bones, tendons, muscle pairs. Healthy choices chart.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook; then the Unit 1 Review.
Exit Ticket	3	Next: Unit 1 Test.

COMMON MISCONCEPTIONS

- Thinking muscles push.

SUPPORT

- Model with two rubber bands and a hinged ruler.

EXTEND

- Track sleep and activity for a week.

STUDENT LESSON 1.8 · WITH ANSWERS

Lesson 1.8: Bones, Muscles, and Healthy Choices

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Do the Investigate activity: Muscle Partners.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–E).

I WONDER

Bend and straighten your arm. What do you feel happening in your upper arm?

An adult skeleton has 206 bones. Bones support the body, protect organs (the skull protects the brain; the ribs protect the heart and lungs), and make blood cells inside. Muscles are attached to bones by tendons. Muscles can only pull, not push, so they work in pairs.

EXPLAIN IT · MUSCLE PAIRS

Bending and straightening your arm

1 To bend: the biceps (front of upper arm) tightens and pulls the forearm up; the triceps relaxes.

2 To straighten: the triceps (back of upper arm) tightens and pulls; the biceps relaxes.

Explanation: Muscles work in pairs because each muscle can only pull.

Healthy choice	Systems it helps
active play about 60 minutes a day	muscular, skeletal, circulatory, respiratory
foods like fruits, vegetables, whole grains, and protein	digestive (and every system that needs nutrients)
9–12 hours of sleep a night for children your age	nervous (and all systems repair during sleep)
drinking water	digestive, circulatory
wearing a helmet and seat belt	skeletal, nervous

INVESTIGATE · MUSCLE PARTNERS

How do muscles work in pairs?

You need: your own arm.

1 Place your hand on your biceps. Slowly bend your arm. What does the muscle do?

2 Move your hand to your triceps. Straighten your arm. What does that muscle do?

Record: a labeled sketch of the upper arm showing which muscle pulls to bend and which pulls to straighten.

★ KEY IDEA

Bones support and protect; muscles pull in pairs to move bones at joints. Healthy habits keep all systems working.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. How many bones in an adult skeleton?

Answer: 206

2. What connects muscle to bone?

Answer: a tendon

3. Why do muscles work in pairs?

Answer: Muscles can only pull, so one pulls to bend and the other pulls to straighten.

Workbook Answer Key · Lesson 1.8

#	Question	Answer
1	How many bones in an adult skeleton?	206
2	What connects muscle to bone?	a tendon
3	Why do muscles work in pairs?	Muscles can only pull, so one pulls to bend and the other pulls to straighten.
4	What do the ribs protect?	the heart and lungs
5	A place where bones meet is a...	joint
6	Muscle that bends the arm	biceps
7	Muscle that straightens the arm	triceps
8	Connects bone to bone	ligament
9	The skull protects the...	brain

#	Question	Answer
10	Write a one-day plan with three healthy choices. Name the system each one helps.	Answers vary.
11	Describe how the skeletal and muscular systems work together to help you walk.	Muscles pull on bones at joints; bones support the body and act as levers.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 1 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Observation or inference: “The cat is sleepy.”	inference
2	In a fair test, how many variables do you change on purpose?	one
3	Smallest unit of life	cell
4	Carries blood back to the heart	vein
5	Tiny vessels where materials pass to cells	capillaries
6	Muscle that bends the arm	biceps
7	Tube from mouth to stomach	esophagus
8	Muscle under the lungs	diaphragm
9	How do the digestive, respiratory, and circulatory systems work together to give a muscle cell energy?	Digestive gives nutrients, respiratory gives oxygen, and the circulatory system carries both to the cell.

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 1 Test — Answer Key

Answers

#	Question	Answer
1	Observation or inference: “The soup is steaming.”	observation
2	Observation or inference: “The soup is too hot to eat.”	inference
3	Plan a fair test: Does a ball roll farther on carpet or on tile? Name what you change, keep the same, and measure.	Change: surface. Same: ball, ramp height, push. Measure: distance rolled.
4	Which list is in order from smallest to largest? A. organ, tissue, cell, system B. cell, tissue, organ, system C. tissue, cell, system, organ	B. cell, tissue, organ, system
5	Which system includes the lungs?	respiratory
6	Which system includes the brain and nerves?	nervous
7	Where do most nutrients pass into the blood?	small intestine
8	What gas do cells need?	oxygen
9	Which blood vessels carry blood away from the heart?	arteries
10	What connects a muscle to a bone?	tendon
11	Why do muscles work in pairs?	Muscles can only pull.
12	Which is chemical digestion? A. chewing B. stomach churning C. stomach acid breaking down food	C. stomach acid breaking down food
13	A pulse of 24 beats in 15 seconds is how many beats per minute?	96
14	Resting breaths: 18/min; after exercise: 38/min. What changed and by how much?	It increased by 20 breaths per minute.
15	Write a CER: Does exercise increase heart rate? Use this data: resting 80, after running 128 beats per minute.	Claim: yes. Evidence: 80 → 128 bpm. Reasoning: muscles need more oxygen, so the heart pumps faster.
16	Choose an activity. Explain how at least three body systems work together during it.	Rubric: three correct systems (1), a correct job for each (1), a clear explanation of how they connect (1).

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; review missed skills.
Below 15	Reteach and retest.

Grade 4 Science

Teacher Edition

A full year of Grade 4 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Teacher Edition holds everything needed to teach: a week-by-week pacing guide, a guided lesson plan for every lesson with timing, discussion prompts, common misconceptions, support and extension ideas, and complete answer keys.

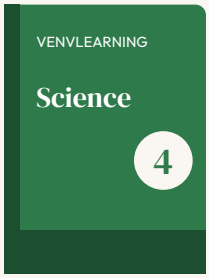
INSIDE THIS YEAR

01 The Human Body

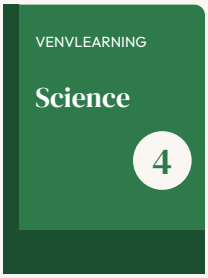
IN THIS BOOK

- Week-by-week pacing guide
- Guided lessons with timing and teaching moves
- Misconceptions, supports, and extensions
- Answer keys and scoring guides for every page

THE COMPLETE SET



Textbook



Workbook



Teacher Edition

You are holding this book

Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.